Realtime Water Quality Indicator (RWQI)

An Engineering Project in Community Service

Phase – II Report

Submitted by

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in partial fulfillment of the requirements for the degree of

Bachelor of Technology

VIT Bhopal University
Bhopal
Madhya Pradhesh

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Bonafide Certificate

Certified that this project report titled **"Realtime Water Quality Indicator"** is the bonafide work of **"Sanjog Gupta 23BCY10105, Akshat Sharma 23BEC10053, Kriti Sapkota 23BCE10142, Anish Bhattacharjee 23BAI10736, Pracheeta Shah 23BCY10155 "** who carried out the project work under my supervision.

This project report (Phase II) is submitted for the Project Viva-Voce examination held on …………..

Supervisor

Comments & Signature (Reviewer 1)

Comments & Signature (Reviewer 2)

Declaration of Originality

We, hereby declare that this report entitled **"Realtime Water Quality Indicator (RWQI)** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

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Acknowledgement

This work has benefited in various ways from several people. Whilst it would be simple to name them all, it would not be easy to thank them enough.

We would like to thank our internal guide **Dr P Narendra Babu** for continually guiding and actively participating in our project, giving valuable suggestions to complete the project work.

We would like to thank all the technical and teaching staff of the college, who extended directly or indirectly all support.
We appreciate the efforts and collaboration of every team member, and thank VIT Bhopal University for providing us with all necessary resources and opportunities. Our heartfelt thanks to everyone who contributed to the success of this engineering project in community service.

Abstract

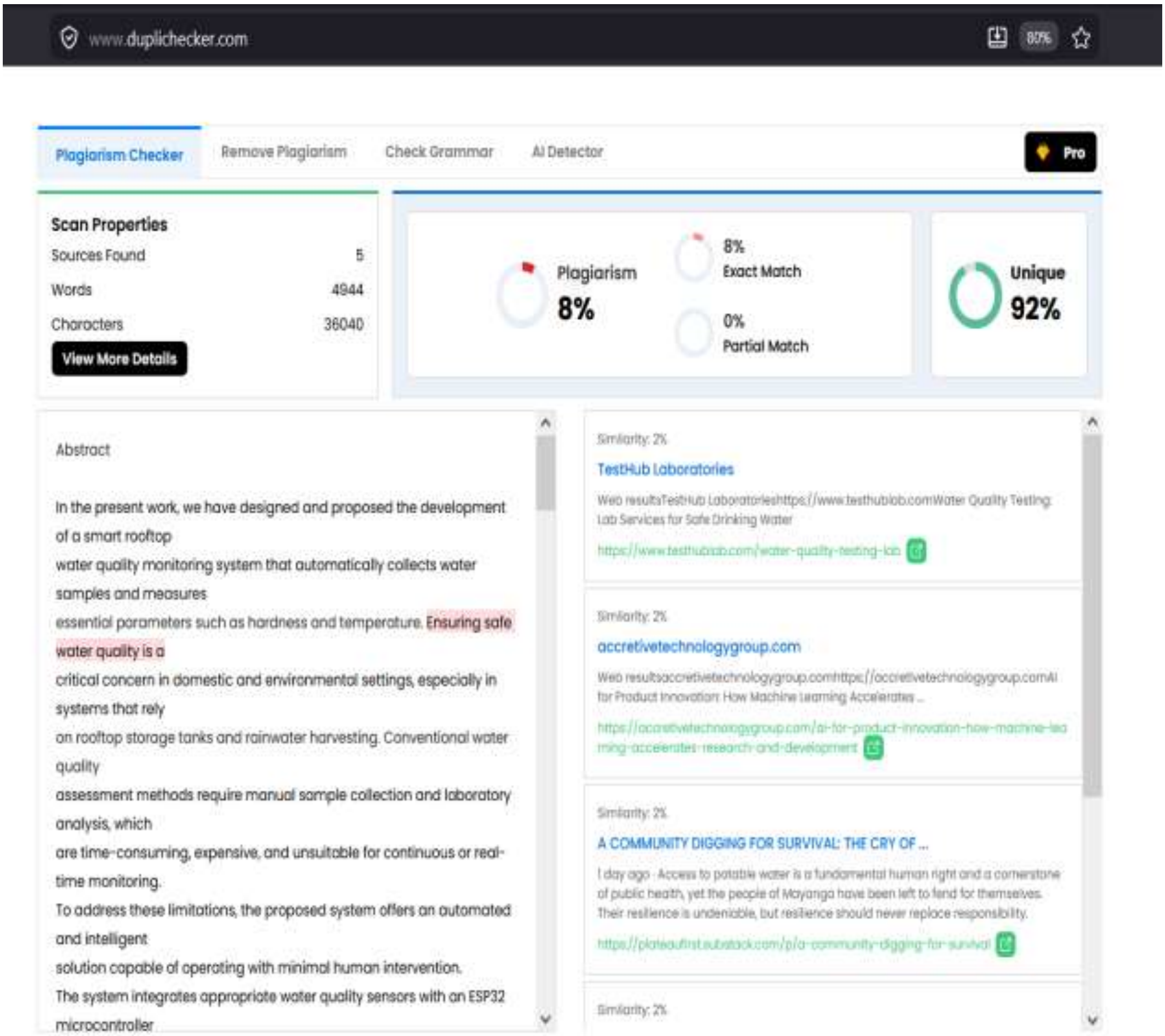
In the present work, we have designed and proposed the development of a smart rooftop water quality monitoring system that automatically collects water samples and measures essential parameters such as hardness and temperature. Ensuring safe water quality is a critical concern in domestic and environmental settings, especially in systems that rely on rooftop storage tanks and rainwater harvesting. Conventional water quality assessment methods require manual sample collection and laboratory analysis, which are time-consuming, expensive, and unsuitable for continuous or real-time monitoring. To address these limitations, the proposed system offers an automated and intelligent solution capable of operating with minimal human intervention.

The system integrates appropriate water quality sensors with an ESP32 microcontroller to acquire, process, and manage the measured data efficiently. The ESP32 acts as the central control unit, handling sensor data acquisition, preliminary processing, and communication. Its built-in Wi-Fi capability enables seamless wireless transmission of real-time water quality data to a user interface, such as a mobile application or web-based dashboard. This allows users to remotely monitor water conditions and take necessary actions if abnormal values are detected.

The proposed system is designed to be compact, cost-effective, and easy to deploy on rooftops or water storage systems. By reducing reliance on manual testing and laboratory procedures, the system promotes continuous monitoring, faster detection of potential issues, and improved decision-making. Overall, this project contributes toward the development of smarter, more accessible, and efficient water management solutions for domestic applications.

Plagiarism Check

Website used: www.duplichecker.com



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1. INTRODUCTION

Access to potable water is a fundamental human right and a cornerstone of public health infrastructure; yet, the World Health Organization (WHO) estimates that over 2 billion people globally utilize a drinking water source contaminated with faeces or industrial effluents. Rapid urbanization and aging municipal infrastructure have exacerbated the issue, introducing contaminants such as heavy metals (specifically Lead and Arsenic), suspended sediments, and microbial pathogens into domestic water supplies. While centralized water treatment plants employ rigorous testing standards, the quality of water often degrades during distribution due to pipe corrosion, back-siphonage, and storage tank contamination. Consequently, the "last mile" of water delivery remains a critical blind spot in public health safety.

The existing paradigm for water quality assurance is predominantly reactive and centralized. Consumers typically rely on sensory perception (smell or visible discoloration) or expensive laboratory analysis, which introduces a significant time delay between contamination events and detection. Commercial handheld TDS meters, while popular, offer a reductionist view of water quality, measuring only electrical conductivity while failing to detect specific toxicity markers like lead (Pb), bacterial turbidity, or pH imbalances. There is, therefore, an urgent engineering imperative to develop a "frugal," autonomous, and comprehensive monitoring solution capable of operating at the residential edge.

This project proposes the design and implementation of an **Realtime Water Quality Indicator (RWQI)**. This IoT-enabled sensor node is engineered to perform real-time, multi-parameter analysis of domestic water. By integrating a modular sensor array with an ESP32 microcontroller, the system quantifies critical physicochemical attributes, including Turbidity (Nephelometric analysis), Total Dissolved Solids (TDS), pH, and Temperature. Furthermore, the project introduces a novel hybrid approach to heavy metal detection by utilizing Computer Vision and Optical Colorimetry. A dedicated camera module analyzes standard reagent strips within a controlled flow cell to detect lead and chlorine, bridging the gap between low-cost digital sensors and complex chemical assays. The ultimate goal is to empower communities with actionable data, aligning with the United Nations Sustainable Development Goal (SDG) 6: Clean Water and Sanitation.

1.1 Motivation

The primary motivation for this project stems from the alarming disparity between water availability and water safety. In many developing regions, water flowing from a tap is not necessarily safe for consumption. The prevalence of waterborne diseases such as cholera, typhoid, and dysentery is directly linked to the lack of real-time monitoring infrastructure. However, beyond biological pathogens, the "invisible threat" of heavy metal contamination—particularly Lead (Pb)—serves as a critical driver for this research. Lead exposure, even at minute concentrations, causes irreversible neurological damage in children and cardiovascular issues in adults. Current solutions for lead detection are either prohibitively expensive (mass spectrometry) or cumbersome (manual chemical kits), leaving the average household vulnerable.

From a technological standpoint, the motivation lies in the democratization of sensor technology. The advent of high-performance, low-cost microcontrollers like the ESP32 and the miniaturization of optical sensors allow for the development of "Lab-on-a-Chip" architectures that were previously impossible outside of industrial settings. We are motivated by the challenge of Sensor Fusion—taking disparate data points (opacity, conductivity, color) and synthesizing them into a clear "Drink/Don't Drink" recommendation for the end-user.

Furthermore, this project is driven by the potential for Community-Level Impact. A single device protects a household, but a network of these devices creates a "Smart Water Grid." If multiple nodes in a neighborhood detect a spike in turbidity or a drop in pH simultaneously, it indicates a systemic failure in the local pipeline rather than a household issue. This capability to crowdsource water quality data can alert municipal authorities to leaks or contamination events faster than traditional methods. Therefore, this project is not merely about building a gadget; it is about engineering a proactive health defense system that leverages the Internet of Things (IoT) to solve a persistent humanitarian crisis.

1.2 Objective

The primary objective of this project is to design, fabricate, and test a low-cost, automated water quality profiling system. The specific technical and operational objectives are as follows:

- **To Design a Modular Flow-Cell Architecture:** To engineer a light-proof, water-tight sampling chamber that isolates incoming water for static analysis, ensuring that optical sensors (turbidity and camera) operate without interference from ambient light or flow turbulence.
- **To Implement Multi-Parameter Sensing:** To integrate and calibrate a suite of sensors to measure:
  - **Physical Parameters:** Temperature (°C), Turbidity (NTU), and Total Dissolved Solids (TDS - ppm).
  - **Chemical Parameters:** pH level (Acidity/Alkalinity).
- **To Develop an Optical Colorimetric Analysis System:** To utilize an ESP32-CAM module and Computer Vision algorithms to analyze chemical reagent strips, specifically for the detection of Lead (Pb) and Residual Chlorine, and to detect general water discoloration (RGB anomalies).
- **To Engineer an IoT-Based Telemetry System:** Use the Wi-Fi capabilities of the ESP32 SoC to transmit sensor data to a cloud dashboard (e.g., Blynk/ThingSpeak) for real-time visualization and historical logging.
- **To Develop an Alerting Mechanism:** To program threshold-based logic that triggers immediate alerts (via mobile notification or local buzzer) when water quality parameters exceed WHO safety guidelines.
- **To Optimize for Cost and Power:** To ensure the prototype is constructed using Commercial Off-The-Shelf (COTS) components, making it replicable and affordable for deployment in low-income communities.

2. Existing Work / Literature Review

1.HydroSense: A Dual-Microcontroller IoT Framework for Real-Time Multi-Parameter Water Quality Monitoring with Edge Processing and Cloud Analytics Abdul Hasib1, A. S. M. Ahsanul Sarkar Akib2, Anish Giri3 arXiv:2601.21595v1 [cs.CV] 29 Jan 2026

The paper *"HydroSense: A Dual-Microcontroller IoT Framework for Real-Time Multi-Parameter Water Quality Monitoring with Edge Processing and Cloud Analytics"* presents an Internet of Things (IoT) based system designed to continuously measure multiple water quality parameters, including pH, dissolved oxygen (DO), temperature, total dissolved solids (TDS), estimated nitrogen, and water level in real time. HydroSense uses a dual-microcontroller architecture where an Arduino Uno handles high-precision analog sensing and an ESP32 manages digital sensing, edge processing, wireless communication, and cloud integration. The system employs advanced calibration and signal-processing techniques to improve accuracy. Tested over 90 days in various environments, HydroSense demonstrated high measurement precision (e.g., ±0.08 pH units, ±0.2 mg/L DO, ±1.9 % TDS) and nearly 100 % reliable cloud data transmission, all while costing only about 300 USD, roughly 85 % cheaper than commercial alternatives.

Motivation:

This work motivates the development of a low-cost, IoT-based water quality monitoring system to overcome the limitations of manual sampling and expensive laboratory testing. It highlights the need for real-time, automated monitoring that can provide accurate water quality data with minimal human intervention.

2.IoT-Based Smart Water Quality Monitoring System
Varsha Lakshmikantha, Anjitha Hiriyannagowda, Akshay Manjunath, Aruna Patted, Jagadeesh Basavaiah, Audre Arlene Anthony
https://www.sciencedirect.com/science/journal/2666285X/2/2

The paper *"IoT-Based Smart Water Quality Monitoring System"* presents a smart monitoring solution that uses IoT technology and sensors to continuously measure key water quality parameters such as temperature, pH, turbidity, and dissolved solids. Sensor data is collected using a microcontroller and transmitted wirelessly to a cloud platform, where it can be monitored in real time by users. The system aims to reduce dependence on manual sampling and laboratory testing by providing an automated, real-time, and cost-effective approach to water quality assessment, making it suitable for domestic and environmental applications.

Motivation:

This work is motivated by the growing need for real-time and continuous water quality monitoring, as traditional testing methods are slow, labor-intensive, and unsuitable for early detection of contamination. The paper emphasizes using IoT to make water monitoring more efficient, accessible, and affordable.

3.Qualitative and quantitative monitoring of drinking water through the use of a smart electronic tongue Arrieta, Álvaro A. ; Márquez, Said ; Mendoza, Jorge Drinking Water Engineering and Science, Volume 15, Issue 2, 2022, pp.25-34

The paper presents a smart electronic tongue system designed to analyze water quality by using a combination of electrochemical sensors and pattern recognition techniques. The system is capable of qualitatively and quantitatively assessing multiple water parameters, including hardness and other chemical characteristics, by interpreting sensor response patterns. The proposed approach demonstrates that sensor arrays, when combined with data processing algorithms, can effectively evaluate water quality without relying on traditional laboratory-based chemical analysis.

Motivation:

The study is motivated by the need to replace complex, time-consuming, and expensive laboratory water testing with a faster and automated sensing approach. It emphasizes developing an intelligent sensor-based system that can efficiently assess multiple water quality parameters in real time.

Comparison Table

Aspect	**HydroSense: Multi-Parameter IoT Framework**	**IoT-Based Smart Water Quality Monitoring System**	**Smart Electronic Tongue for Water Analysis**
Main Objective	Real-time, low-cost multi-parameter water quality monitoring using IoT	Continuous monitoring of water quality using IoT and cloud platforms	Automated chemical analysis of water using sensor arrays
Technology Used	Dual microcontrollers (Arduino + ESP32), IoT, cloud analytics	Microcontroller-based sensors with IoT connectivity	Electrochemical sensor array with pattern recognition
Parameters Measured	pH, temperature, dissolved oxygen, TDS, water level, nitrogen	Temperature, pH, turbidity, TDS	Hardness and other chemical properties
Data Handling	Edge processing + cloud-based real-time visualization	Wireless transmission to cloud for user monitoring	Local data processing and classification

3. Topic of the work

3.1 System Design / Architecture

The Realtime Water Quality Indicator (RWQI) is designed as an embedded system comprising three functional layers: The Hydraulic Layer, the Hardware Layer (Electronics), and the Software Layer (Processing & IoT).

The system consists of a rooftop water pipe inlet at houses connected to a controlled sampling chamber (flow cell). Water flow is directed into a small enclosed reservoir through electronically controlled valves. Inside this chamber, multiple sensors are positioned strategically to ensure accurate and stable readings.

At the core lies the ESP32-WROOM-32 microcontroller, which acts as the central processing and communication unit. It collects raw analog and digital signals from sensors, processes them, performs calibration and compensation, and transmits the final data to a cloud dashboard via Wi-Fi.

The architecture integrates:

- TDS Sensor (conductivity-based measurement)
- Turbidity Sensor (optical scattering principle)
- pH Sensor (electrochemical measurement)
- DS18B20 Temperature Sensor (digital 1-Wire protocol)
- ESP32-CAM module (for optical colorimetric analysis of reagent strips)

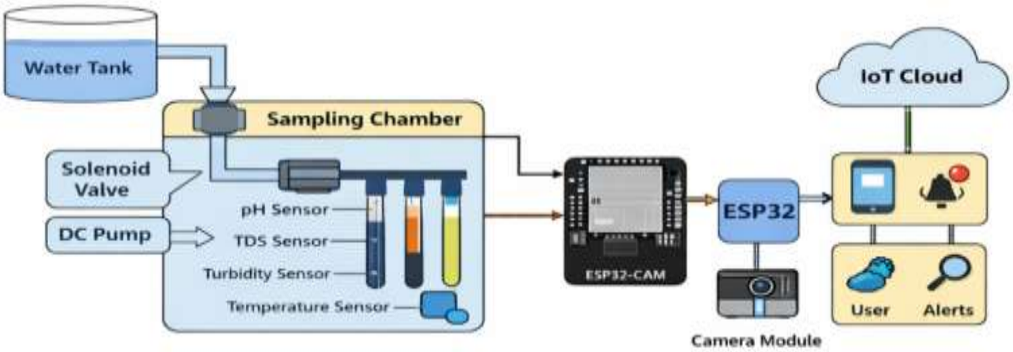
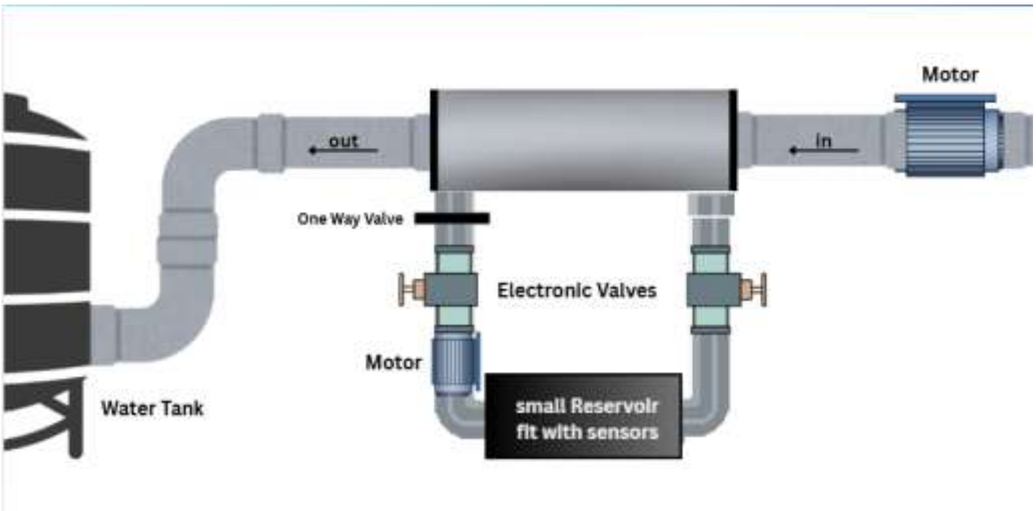


Fig. 1. Architecture of IWQMS



3.1.1 Hydraulic Subsystem

The hydraulic design ensures:

- Controlled water sampling
- Isolation from ambient light
- Minimal turbulence for stable optical readings
- Automated flushing to prevent residue accumulation

A one-way valve prevents backflow contamination. The sealed chamber design reduces measurement errors caused by air bubbles or external light interference.

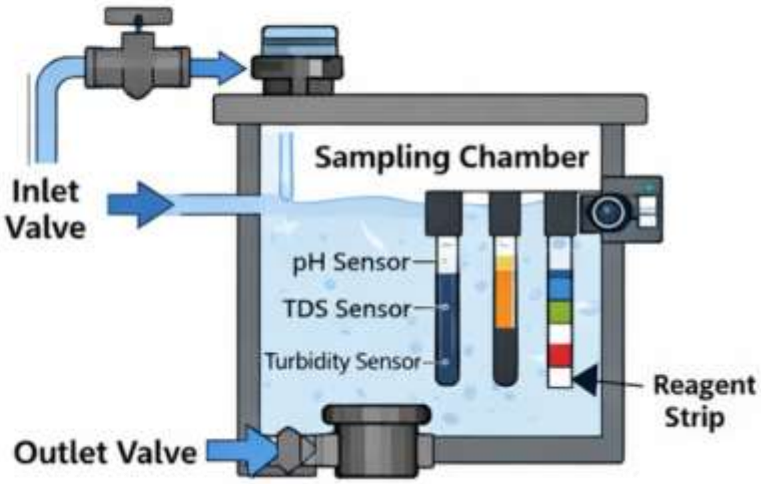


Fig. 2. Flow Cell and Sampling Mechanism

3.1.2 Electronic Architecture

The ESP32 interfaces with:

- Analog sensors via ADC pins
- Digital temperature sensor via One-Wire protocol
- Camera module via dedicated GPIO and serial communication
- Relay/Motor driver for actuation

The ESP32's dual-core processor allows simultaneous sensor acquisition and wireless data transmission.

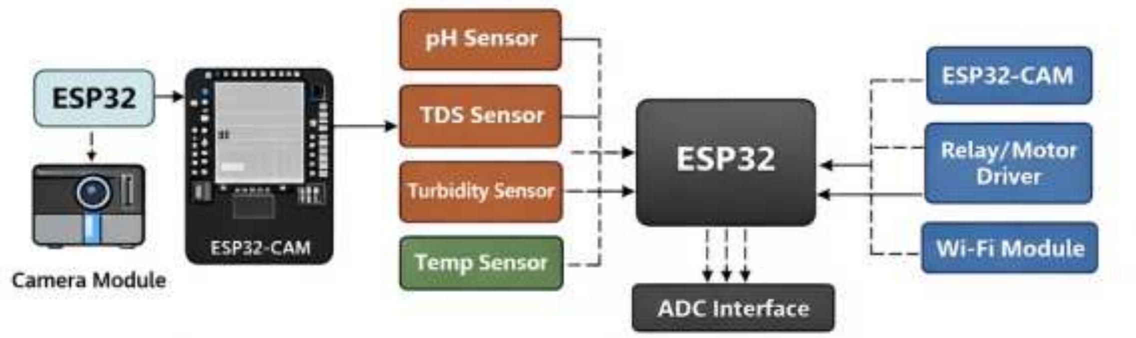


Fig. 3. Hardware Block Diagram

3.1.2 Software Architecture

The software architecture is developed using Tinkercad.

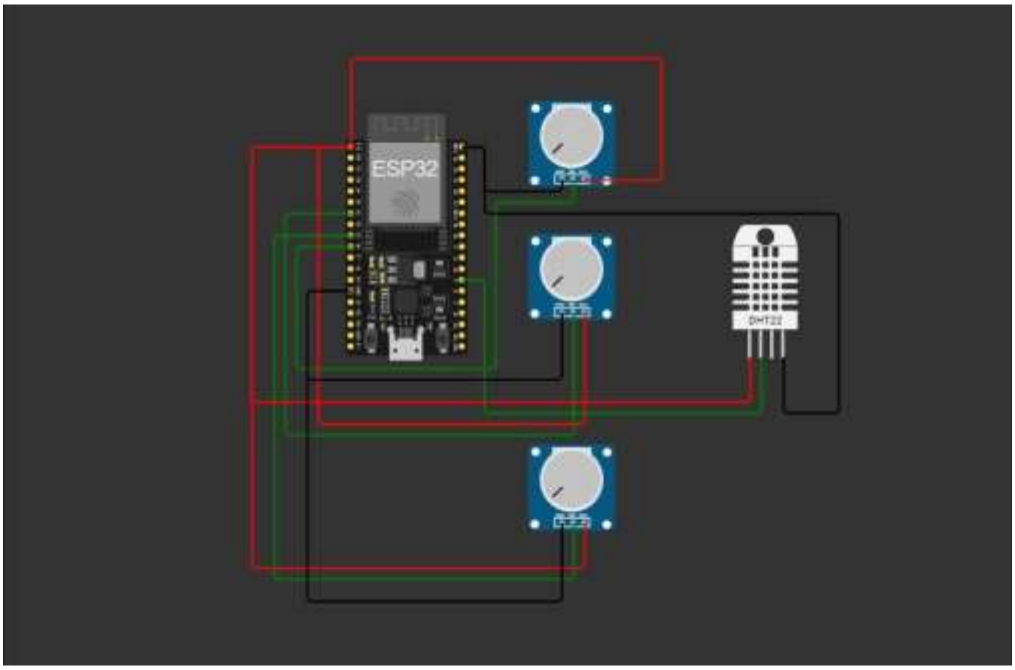
Before implementing the hardware, the system was first designed and visualized using circuit design software to obtain a clear understanding of component connections and data flow.

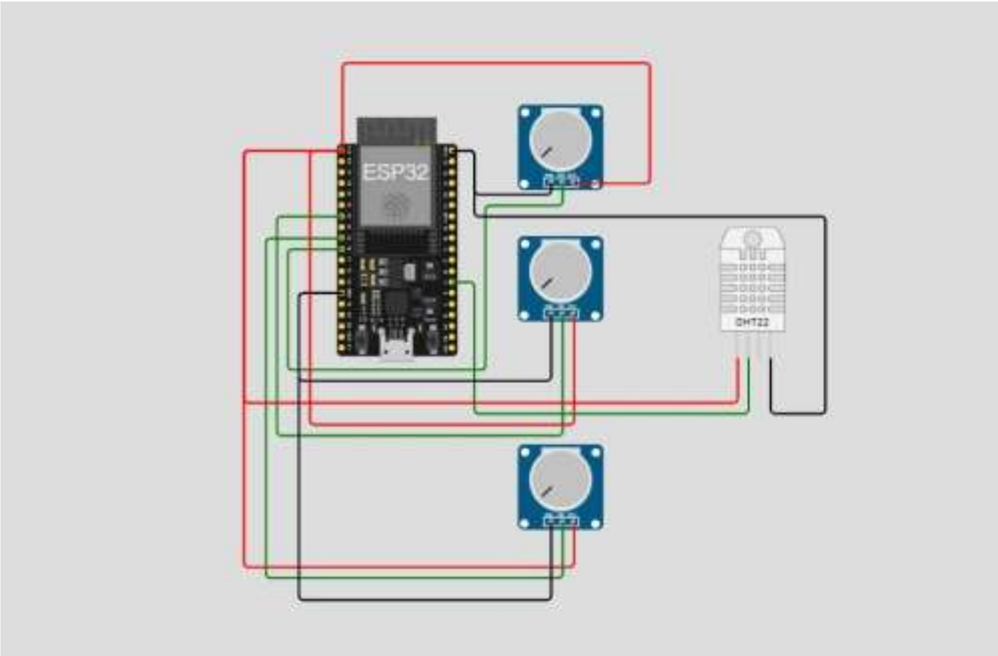
As shown in the model, the ESP32 acts as the central processing unit, interfacing with multiple sensors through its GPIO pins. Each sensor module representing pH, TDS, turbidity, and water flow is connected using three lines

- power (VCC)
- ground (GND)
- and signal pins

ensuring proper data transmission to the microcontroller. The structured wiring layout in the design helps in identifying pin configurations, avoiding connection errors, and optimizing the circuit before physical implementation.

This software-based modeling approach allowed efficient planning of sensor integration, power distribution, and signal routing, ultimately reducing hardware errors and improving system reliability during real-time deployment.





3.2 Working Principle

The proposed smart rooftop water quality monitoring system operates on the principle of real-time sensor-based data acquisition and wireless communication using an embedded microcontroller. Water collected from the rooftop storage system is directed toward the sensing unit, where appropriate sensors measure essential parameters such as hardness and temperature.

The temperature sensor measures the thermal condition of the water, while the hardness sensor estimates the mineral content present in the water sample. These sensors generate electrical signals proportional to the measured values. The ESP32 microcontroller receives these signals, processes the data through programmed logic, and converts them into readable parameter values.

Once processed, the ESP32 uses its built-in Wi-Fi module to transmit the measured data to a user interface, such as a mobile application or web dashboard. This allows users to monitor water quality in real time without manual testing. If abnormal values are detected, the system can be configured to generate alerts.

Thus, the system works by combining sensor measurement, microcontroller-based processing, and IoT-enabled wireless communication to provide continuous and automated water quality monitoring.

3.3 Testing and Validation

We have considered multiple values received by the sensors and have created an observation table to understand and test the system that has been built.

3.3.1 Purpose of Testing

The purpose of testing is to verify that the developed system accurately measures water quality parameters and correctly classifies water for household use.

3.3.2 Testing Setup

The system was tested using the following components:

- ESP32 microcontroller
- pH sensor
- TDS sensor
- Turbidity sensor
- Temperature sensor
- Humidity sensor

Water samples were collected and tested under controlled conditions.

3.3.3 Testing Methodology

- Sensors were calibrated before testing
- The whole software system was integrated with ESP32.
- Each sample was tested **three times**
- Average value was recorded to reduce error
- Readings were compared with predefined standard ranges
- System output (Good / Acceptable / Poor) was verified manually

3.3.4 Standardisation of Range

Standard Range for Domestic Use

Parameter	Ideal	Acceptable	Poor
Temperature	**20–35°C**	**35–40°C**	**>40°C**
pH	**6.0–9.0**	**5.5–6.0 / 9.0–9.5**	**<5.5 or >9.5**
TDS	**<500 ppm**	**500–1000 ppm**	**>1000 ppm**
Turbidity	**0–5 NTU**	**5–10 NTU**	**>10 NTU**
Humidity	**30–80%**	**80–90%**	**>90% / <30%**

Basis of Standardisation

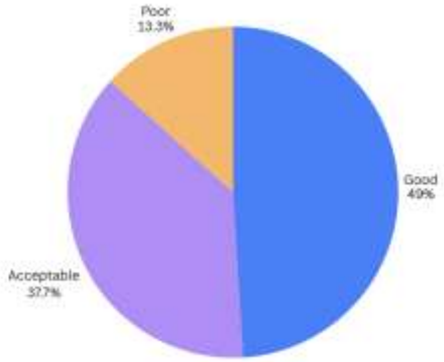
- **Temperature**: Affects usability and comfort in household tasks
- **pH**: Indicates acidity/alkalinity; extreme values can damage pipes and surfaces
- **TDS**: Represents dissolved impurities affecting cleaning efficiency
- **Turbidity**: Indicates suspended particles and water clarity
- **Humidity**: Helps understand environmental conditions affecting water storage

3.3.5 Observation and Analysis

Observation Table

Sample	Temp (°C)	pH	TDS (ppm)	Turbidity (NTU)	Humidity (%)	Status
1	32	7.2	220	1.0	55	Good
2	26	7.3	200	1.2	57	Good
3	27	7.1	240	1.5	57	Good
4	28	6.9	240	2.0	58	Good
5	28	7.5	300	2.5	59	Good
6	32	7.0	400	3.0	60	Good
7	31	6.8	350	3.5	62	Good
8	35	6.7	400	4.0	62	Good
9	43	6.6	450	4.5	58	Good
10	34	6.5	770	5.0	64	Good
11	35	6.4	550	5.5	55	Acceptable
12	36	6.3	600	6.0	66	Acceptable
13	38	6.2	650	6.5	67	Acceptable

14	**40**	**6.1**	**720**	**7.0**	**68**	**Acceptable**
15	**30**	**6.0**	**750**	**7.5**	**69**	**Acceptable**
16	**40**	**5.9**	**780**	**8.0**	**58**	**Acceptable**
17	**41**	**5.8**	**850**	**8.5**	**62**	**Acceptable**
18	**42**	**5.7**	**900**	**9.0**	**72**	**Acceptable**
19	**38**	**5.6**	**820**	**9.5**	**73**	**Acceptable**
20	**40**	**5.5**	**1000**	**10.0**	**74**	**Acceptable**
21	**60**	**5.4**	**1050**	**10.5**	**75**	**Poor**
22	**41**	**5.3**	**1100**	**11.0**	**76**	**Poor**
23	**52**	**5.2**	**1150**	**11.5**	**77**	**Poor**
24	**43**	**5.1**	**1200**	**12.0**	**78**	**Poor**
25	**34**	**7.8**	**300**	**2.0**	**60**	**Good**



Out of the 25 samples tested:

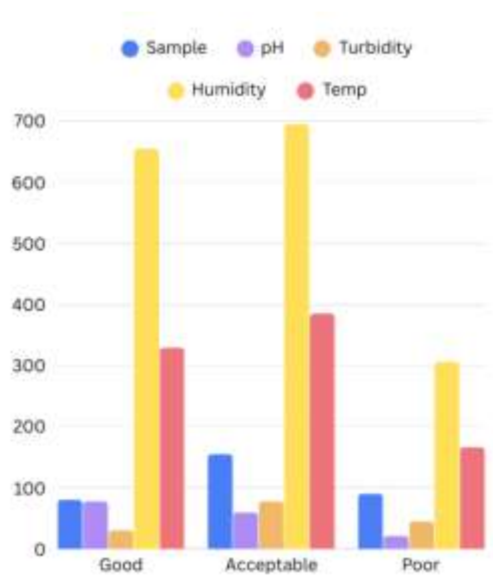
Good = 11 samples (49%)

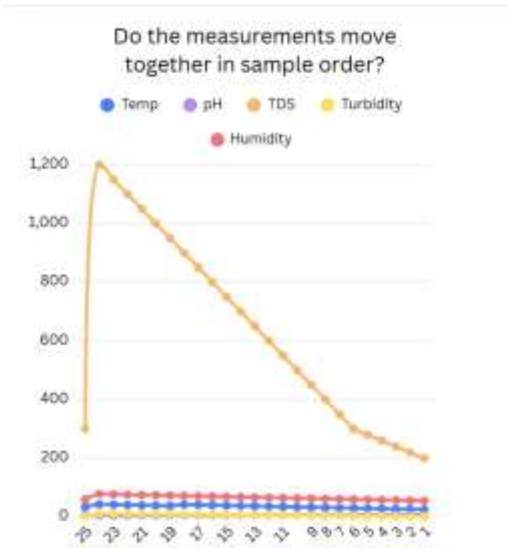
Acceptable = 10 samples (37.7%)

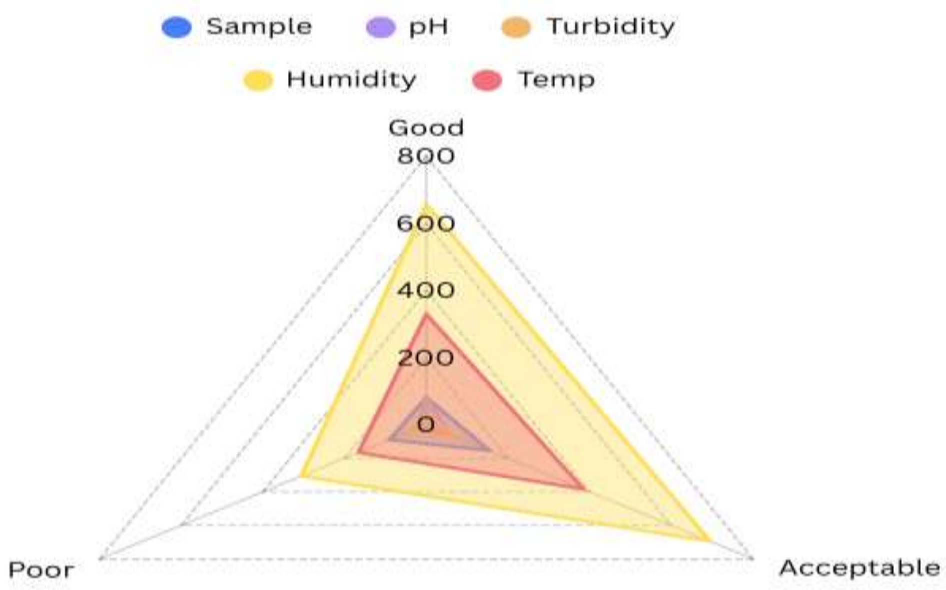
Poor = 4 samples (13.3%)

Graphical Analysis

Comparison of all 5 properties together

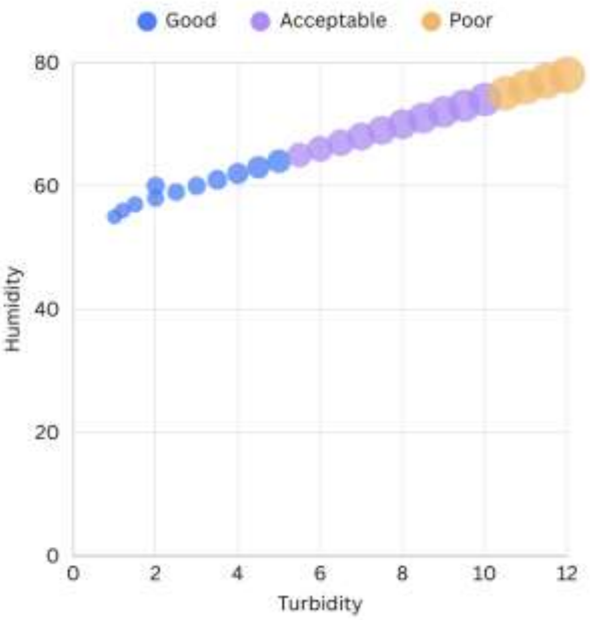




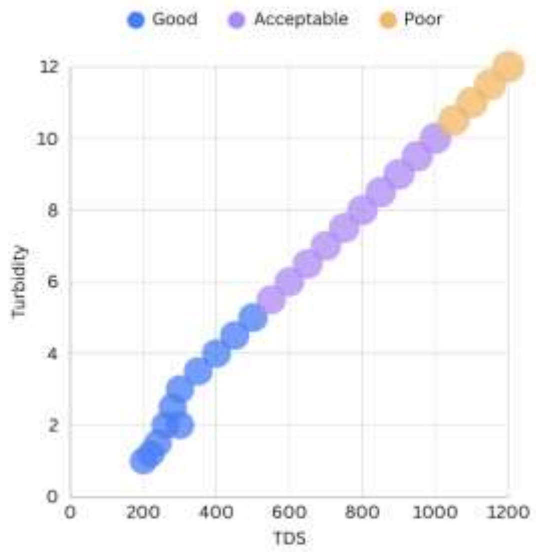


Correlation Analysis

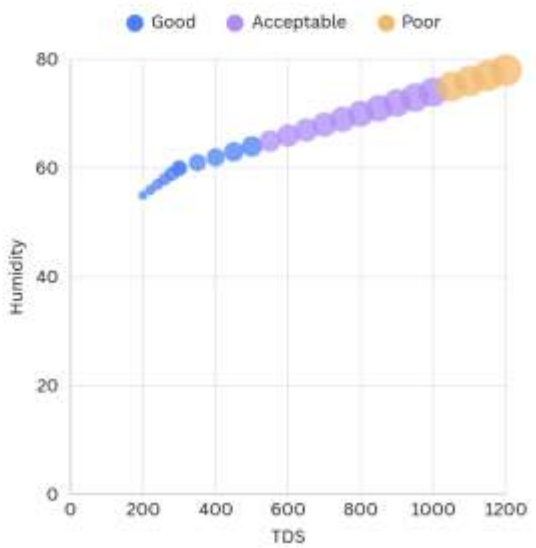
1)Humidity VS Turbidity



2) TDS VS Turbidity



3) Humidity VS TDS

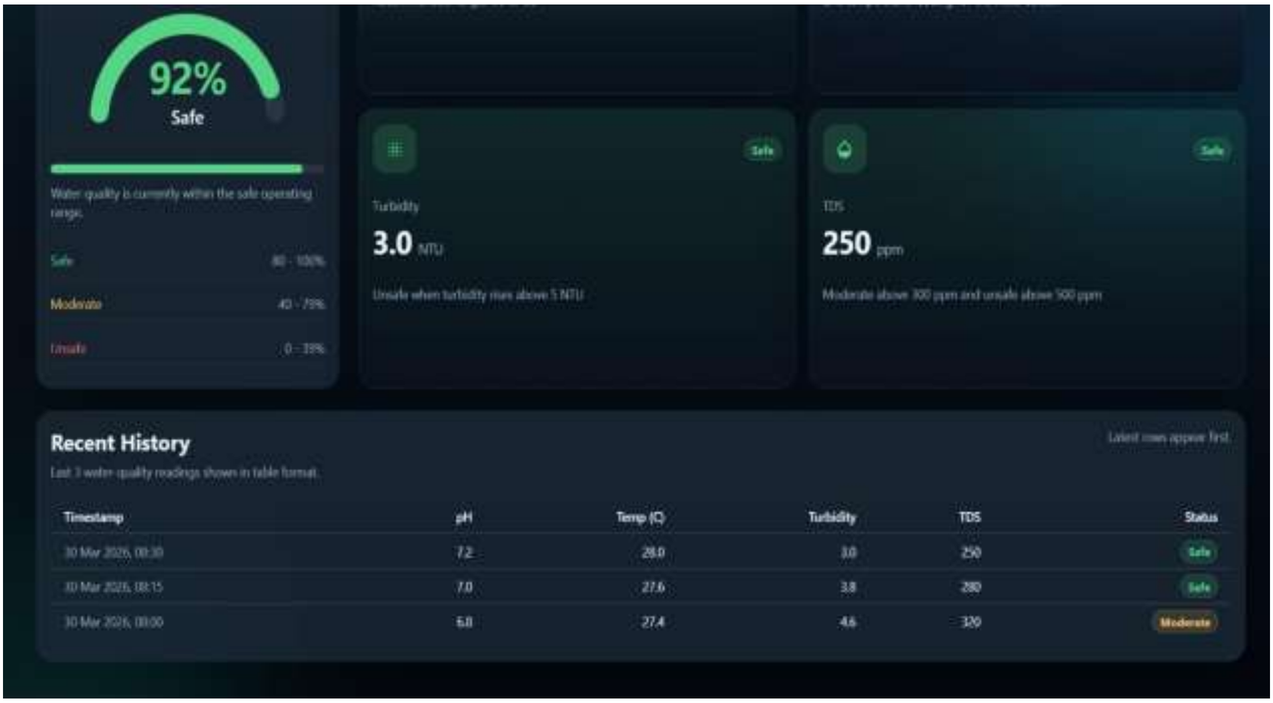


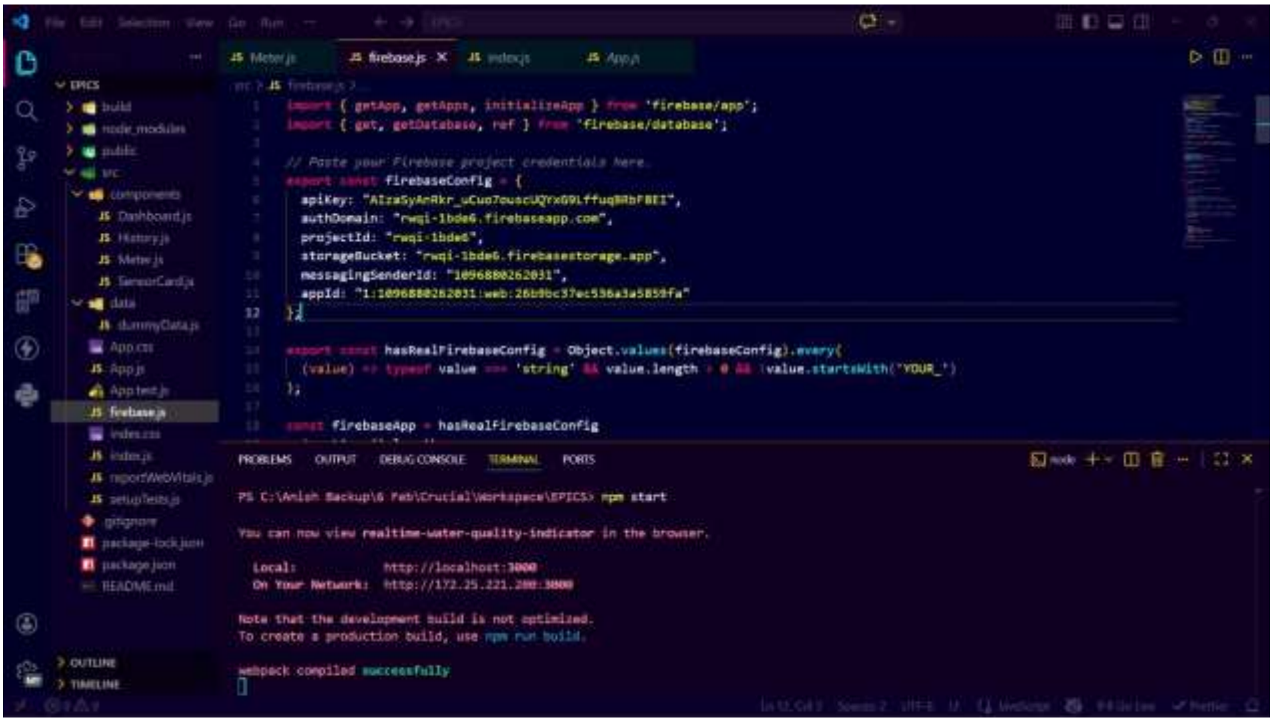
3.4 Software Components

We have designed a realtime website using React for the user to view the data collected by the model.

The software component is responsible for data acquisition, processing, and visualization. The ESP32 microcontroller continuously collects readings from the pH, TDS, turbidity, and water flow sensors and processes this data in real time. Using its built-in Wi-Fi capability, the ESP32 transmits the sensor data to a web-based platform through HTTP protocols.

On the backend, the system stores and organizes incoming data, ensuring reliability and scalability. The frontend website is designed to display the readings in an intuitive and user-friendly manner, often using live updates and timestamps to help users monitor water quality parameters effectively.


Realtime Water Quality Indicator (RWQI)
Responsive monitoring dashboard for engineering projects with Firebase-ready live data support.
All monitored water-quality values are within the safe operating range.
30 Mar 2026, 20:28
Water Quality Meter
92%
Safe
7.2 pH
28.0
Stable

92%
Safe
Turbidity
3.0 NTU
TDS
250 ppm
Recent History
Timestamp
pH
Temp (C)
Turbidity
TDS
Status
Moderate


Meter.js
firebase.js
index.js
App.js
EPICS
build
node_modules
public
src
components
Dashboard.js
History.js
Meter.js
SensorCard.js
data
dummyData.js
App.css
App.js
App.test.js
firebase.js
index.css
index.js
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json
README.md
OUTLINE
TIMELINE
import { getApp, getApps, initializeApp } from 'firebase/app';
import { get, getDatabase, ref } from 'firebase/database';
// Paste your Firebase project credentials here.
export const firebaseConfig = {
authDomain: "rwqi-1bde6.firebaseapp.com",
projectId: "rwqi-1bde6",
storageBucket: "rwqi-1bde6.firebasestorage.app",
messagingSenderId: "1096880262031",
};
export const hasRealFirebaseConfig = Object.values(firebaseConfig).every(
(value) => typeof value === 'string' && value.length > 0 && !value.startsWith('YOUR_')
);
const firebaseApp = hasRealFirebaseConfig
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Anish Backup\6 Feb\Crucial\Workspace\EPICS> npm start
You can now view realtime-water-quality-indicator in the browser.
Local: http://localhost:3000
On Your Network: http://172.25.221.200:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully

3.5 Results and Discussion

The preliminary implementation of the proposed smart rooftop water quality monitoring system focused on sensor integration, ESP32 configuration, and basic data transmission. During testing, the temperature sensor was able to successfully detect and display variations in water temperature under different environmental conditions. The ESP32 microcontroller effectively processed sensor data and transmitted readings wirelessly to the designated interface, demonstrating stable communication using its built-in Wi-Fi capability.

Initial observations indicated that the system responded to changes in input conditions and was capable of producing consistent output values within expected ranges. However, since the system is still under development, calibration and long-term performance validation are ongoing. The hardness measurement module requires further fine-tuning to improve measurement stability and accuracy under varying water compositions.

The results demonstrate that the proposed architecture is functional at a prototype level and capable of real-time monitoring. Further improvements in sensor calibration, enclosure design, and long-duration testing will enhance system reliability and performance before full-scale deployment.

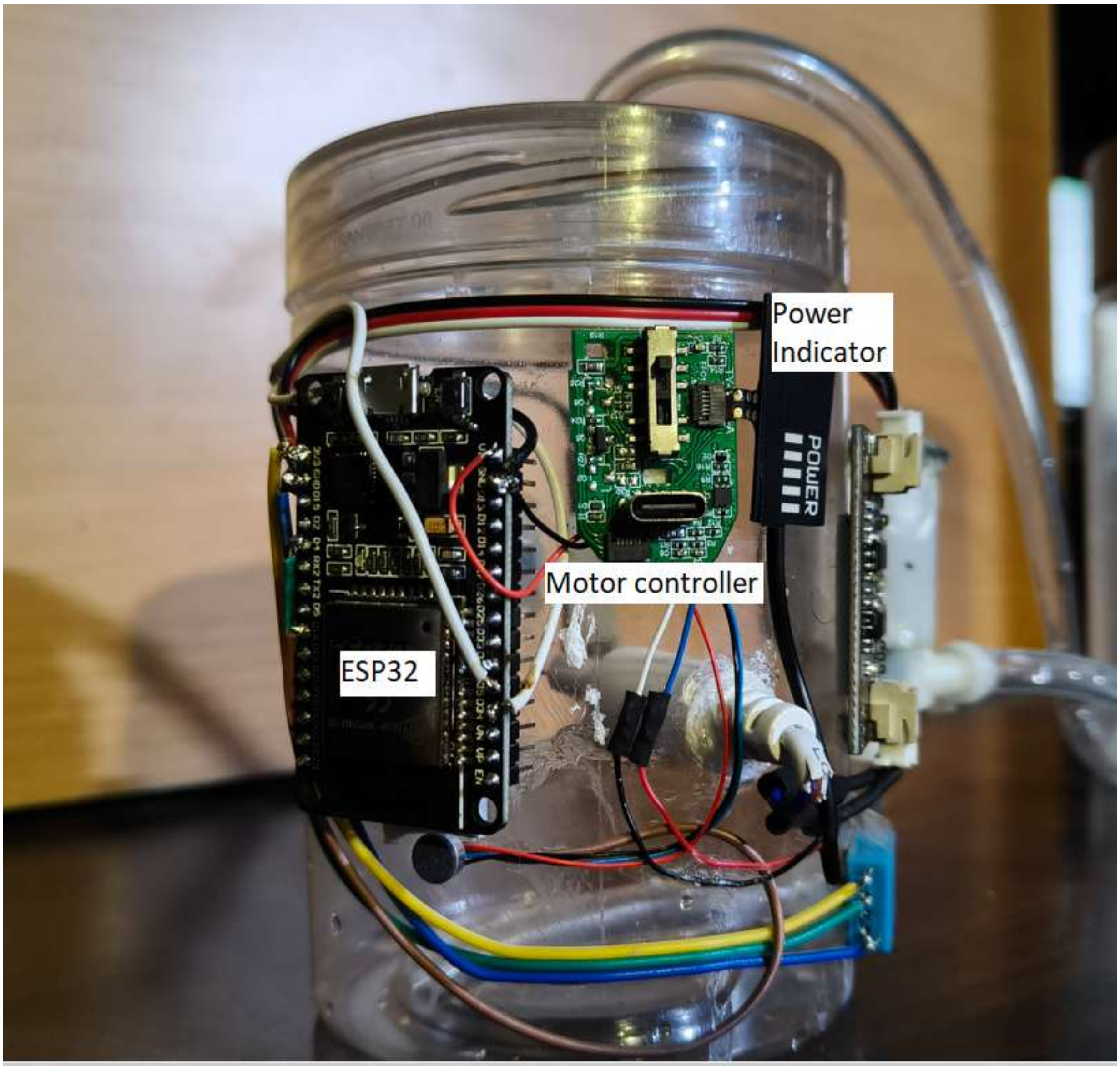
3.6 Contribution

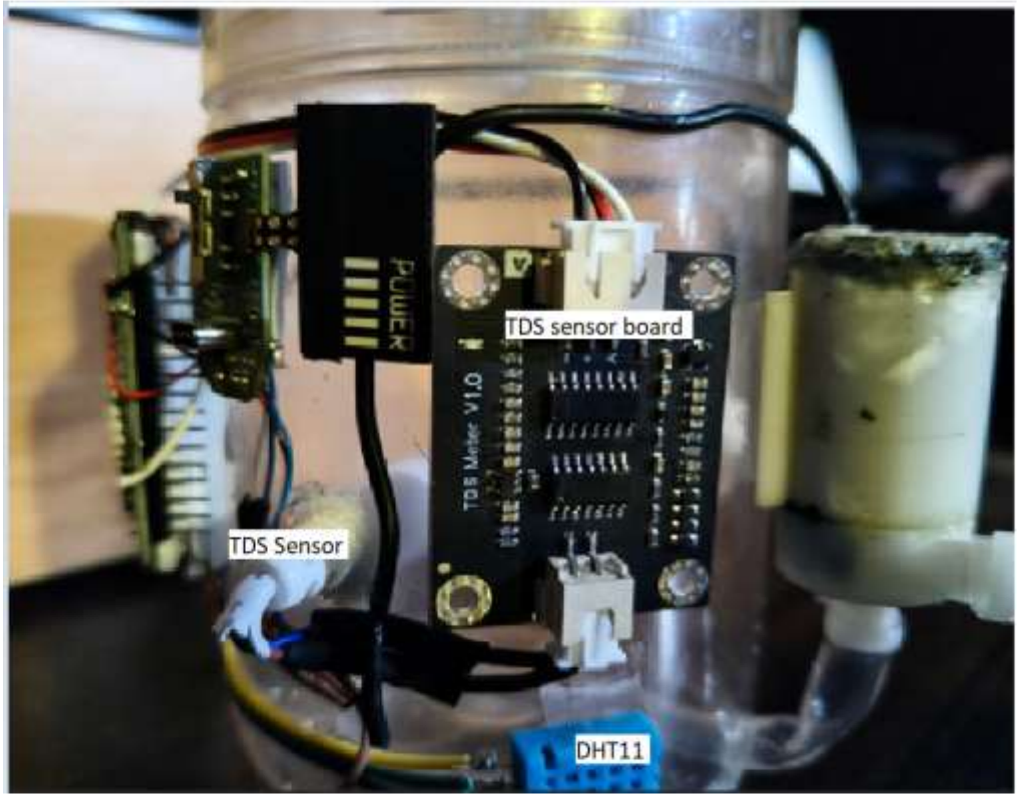
Anish Bhattacharjee

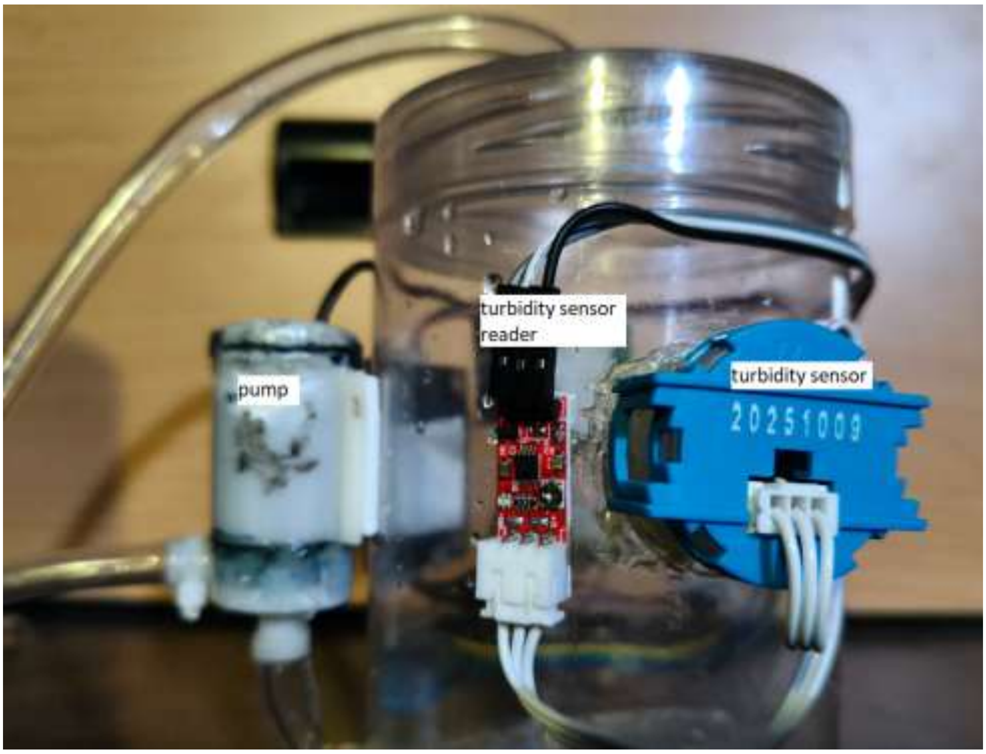
Anish was responsible for handling the software aspect of the project, including the design and development of the web application.

He developed the website to display real-time sensor data received from the ESP32, ensuring that the interface was user-friendly and easy to interpret. His work involved integrating data communication between the hardware and the web platform, organizing the data for proper visualization, and implementing features such as live updates and graphical representation of parameters. This contribution played a key role in enabling effective monitoring and analysis of water quality data.

3.7 PICTURES OF THE HARDWARE




TDS sensor board
TDS Sensor
DHT11


turbidity sensor
reader
pump
turbidity sensor
20251009

4. CONCLUSION:

The proposed smart rooftop water quality monitoring system presents an effective and practical solution for continuous domestic water quality assessment. The system successfully integrates water quality sensors with the ESP32 microcontroller to measure essential parameters such as hardness and temperature. By utilizing the built-in Wi-Fi capability of the ESP32, the collected data is transmitted in real time to a user interface, enabling remote monitoring without manual intervention. This reduces dependency on traditional laboratory-based testing methods, which are often time-consuming and expensive. The project demonstrates how IoT-based embedded systems can be applied to everyday problems, making water quality monitoring more accessible, efficient, and user-friendly for residential applications.

The system is designed to be compact, cost-effective, and easy to install on rooftops or near domestic water storage systems. Its automated operation allows users to continuously monitor water conditions and respond promptly if abnormal values are detected. This contributes to improved awareness and proactive management of water quality at the household level.

In the future, the system can be enhanced by incorporating additional water quality parameters such as pH, turbidity, total dissolved solids (TDS), and dissolved oxygen for more comprehensive analysis. Advanced data analytics and cloud integration can be implemented to track long-term trends and generate alerts for potential contamination. With further development and optimization, the device can be deployed widely in residential areas, enabling installation in multiple homes to monitor the water quality of a specific locality. Such large-scale implementation can help identify regional water quality patterns, support data-driven decision-making, and contribute toward smarter, safer, and more sustainable water management systems.

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6. Biodata with Picture:

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Anish Bhattacharjee is a technically proficient Computer Science Engineering student at VIT Bhopal University, specializing in Artificial Intelligence and Machine Learning. He possesses a versatile skill set spanning C++, Java, Python, and SQL, complemented by strong capabilities in full-stack web development using ReactJS. His portfolio highlights a talent for applying AI to practical solutions, demonstrated through projects like "Agro_World," a predictive agricultural model, and an AI Super Resolution platform for image enhancement. With additional expertise in AWS and IoT systems, Anish is a creative problem-solver dedicated to engineering scalable, data-driven applications and continuously expanding his technical horizons.